

Slope and Rate of Change

Answer Key

Grade 8 / Pre-Algebra Expressions and Equations

Quick Answers

1. 2	5. 18, 6	9. 2, 4
2. 1	6. 14, —	10. 5, —
3. 4, 19	7. -2	11. -2
4. 3	8. -5, 10	12. -5, -2, 3, 6

Curriculum

Level: Grade 8 / Pre-Algebra Expressions and Equations8.EE.B.5, 8.EE.B.6 – *problems 1, 2, 3, 4, 6, 7, 8, 11, 12*8.F.A.1–8.F.B.5 – *problems 5, 9, 10**Difficulty 1–5 of 5 across 19 tagged checks.*

Common wrong answers

3. *If they got 33: multiplied the whole expression by the miles, charging the fixed fee once per mile*
11. *If they got 2: subtracted the coordinates in opposite orders, top and bottom*
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1. Slope through (1, 2) and (4, 8).

Step 1. Two points and no context, so the slope formula applies directly — label one point (x_1, y_1) and the other (x_2, y_2) and keep that labelling for both subtractions.

Step 2. $m = \frac{8-2}{4-1} = \frac{6}{3}$.

Step 3. $m = \boxed{2}$: y goes up 2 for every 1 that x goes up.

2. Line B on Figure 1.

Step 1. Read two points the line passes *exactly* through — a point read between grid lines is a guess, and slope from a guess is a guess.

Step 2. Line B runs through the origin $(0, 0)$ and through $(8, 8)$, so $m = \frac{8-0}{8-0}$.

Step 3. $m = \boxed{1}$.

Teaching note: a slope of 1 is the 45° line. It is worth naming, because students often report the slope of a diagonal as “up and across”, which is a description, not a number.

3. Taxi fare $C = 4d + 7$.

(a) **Step 1.** The equation is already in $y = mx + b$ form, so the rate of change is the coefficient of d — the fee 7 is charged once and never changes, so it cannot be part of a rate.

Step 2. Checking it against two points: at $d = 0$ the cost is 7 and at $d = 1$ the cost is 11, so $m = \frac{11 - 7}{1 - 0}$.

Step 3. The rate is $\boxed{4}$ dollars per mile.

(b) **Step 4.** Substitute $d = 3$: $C = 4(3) + 7 = 12 + 7$.

Step 5. A three-mile ride costs $\boxed{19}$ dollars.

Common wrong answer: 33 — the student computed $(4 + 7) \times 3$, charging the flat pick-up fee once for every mile.

4. Line A on Figure 1.

Step 1. Again pick two lattice points; the graph marks $(0, 2)$ and $(4, 14)$ on the line.

Step 2. $m = \frac{14 - 2}{4 - 0} = \frac{12}{4}$.

Step 3. The rate of change is $\boxed{3}$.

Teaching note: $(0, 2)$ is the y -intercept, so Line A is $y = 3x + 2$. Naming the equation of a line the student has just measured is what ties the graph form to the equation form.

5. Water in the tub.

(a) **Step 1.** “How many litres run in” asks for a change in the output, not a rate, so subtract the two litre readings and stop there.

Step 2. At minute 8 there are 50 litres and at minute 5 there are 32, so the change is $50 - 32 = \boxed{18}$ litres.

(b) **Step 3.** A rate is that change divided by the change in time. Using minutes 2 and 5:
 $m = \frac{32 - 14}{5 - 2} = \frac{18}{3}$.

Step 4. The rate is $\boxed{6}$ litres per minute.

Check it is really constant: from minute 5 to minute 8 the same calculation gives $\frac{18}{3} = 6$ again. Equal rates over every interval is what makes the table linear.

6. Comparing two streaming plans.

(a) **Step 1.** Plan B is a table, so its rate comes from two rows: $m = \frac{84 - 42}{6 - 3} = \frac{42}{3}$.

Step 2. Plan B costs $\boxed{14}$ dollars per hour.

(b) **Step 3.** Plan A is $y = 12x$, which is already $y = mx$, so its rate is 12 dollars per hour. Now both rates are in the same units and the numbers can be compared directly.

Step 4. 12 is less than 14, so Plan A’s rate $\boxed{<}$ Plan B’s rate — Plan A is the cheaper plan per hour.

Listen for: “Plan B is bigger because the numbers in the table are bigger.” The table’s numbers are totals for different numbers of hours; only the rate is comparable.

7. Line C on Figure 1.

Step 1. This line falls from left to right, so expect a negative answer before computing anything — that expectation is the check on the sign.

Step 2. The line passes through $(0, 16)$ and $(8, 0)$, so $m = \frac{0 - 16}{8 - 0} = \frac{-16}{8}$.

Step 3. The rate of change is $\boxed{-2}$.

Step 4. What the sign means: y goes *down* by 2 for every 1 that x goes up. The steepness is 2; the direction is the minus sign.

8. Leaking barrel $L = -5x + 40$.

(a) **Step 1.** The equation is in $y = mx + b$ form, so the rate is the coefficient of x ; the 40 is how much was in the barrel at the start.

Step 2. The rate of change is $\boxed{-5}$ litres per minute — the barrel loses 5 litres each minute.

(b) **Step 3.** Substitute $x = 6$: $L = -5(6) + 40 = -30 + 40$.

Step 4. After six minutes $\boxed{10}$ litres are left.

Worth asking: at what time does the barrel run dry? Setting $-5x + 40 = 0$ gives $x = 8$ minutes, which is where this rule stops describing the barrel.

9. A table that is not linear.

(a) **Step 1.** A single rate only describes the whole table if it is the same over every interval, so compute it interval by interval rather than once.

Step 2. From $x = 0$ to $x = 2$: $m = \frac{4 - 0}{2 - 0} = \boxed{2}$.

(b) **Step 3.** From $x = 2$ to $x = 4$: $m = \frac{12 - 4}{4 - 2} = \frac{8}{2} = \boxed{4}$.

Step 4. The two rates are different, so the points do not lie on one straight line: between the first pair the graph climbs 2 per step, and between the second pair it climbs 4 per step. A linear relationship has one rate everywhere.

Grading note: accept any wording that says the rate changed. Do not accept “it is not linear because the numbers are not even” — evenness is not the test.

10. Mara and Ben.

(a) **Step 1.** Mara’s data is a table, so her speed is the change in metres over the change in seconds: $m = \frac{25 - 10}{5 - 2} = \frac{15}{3}$.

Step 2. Mara runs at $\boxed{5}$ metres per second.

(b) **Step 3.** Ben’s rule $y = 4.5x$ is already $y = mx$, so his speed is 4.5 metres per second. Both are now in metres per second and can be compared.

Step 4. 5 is greater than 4.5, so Mara’s speed $\boxed{>}$ Ben’s speed.

Teaching note: the whole point of this problem is that a table and an equation are not comparable until both have been reduced to a rate in the same units. Doing that reduction is the skill; the comparison afterwards is easy.

11. Slope through $(-3, 5)$ and $(2, -5)$.

Step 1. With negative coordinates the danger is subtracting in opposite orders top and bottom, so fix the labelling first: $(x_1, y_1) = (-3, 5)$ and $(x_2, y_2) = (2, -5)$.

Step 2. $m = \frac{-5 - 5}{2 - (-3)} = \frac{-10}{5}.$

Step 3. $m = \boxed{-2}.$

Common wrong answer: 2 — the numerator was taken as $5 - (-5) = 10$ while the denominator stayed $2 - (-3) = 5$, mixing the two orders. Both subtractions must start from the same point.

12. Synthesis: ordering four rates.

Step 1. Collect the four rates found earlier on the sheet. Line A (Problem 4) is 3; Line C (Problem 7) is -2 ; the tub table (Problem 5) is 6; the barrel equation (Problem 8) is -5 .

Step 2. Order them on a number line, not by how big they look. The two negatives come first, and -5 is further left than -2 .

Step 3. Least to greatest: $\boxed{-5, -2, 3, 6}.$

Step 4. (*Second part: open response — flagged for manual review; read the student's sentence.*) Any of the three answers can earn full credit if the reason fits the representation:

- *equation* — the rate is simply the number in front of x , with no computation at all;
- *table* — one subtraction and one division, and you can check the rate is constant by repeating it on another pair of rows;
- *graph* — you can see the direction and the steepness at a glance, though you must land on lattice points to get an exact value.

Grading note: require a reason tied to how the rate is obtained. “The graph, because I like graphs” is not a justification.

Listen for: a student who orders the rates as 2, 3, 5, 6 by steepness, ignoring sign. A rate of -5 is the fastest *fall*, and it is still the smallest number.